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**EX NO: 9**

**SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES  
USING MATLAB**

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**OBJECTIVES**

- i) To implement convolutional encoding for a given binary input sequence using a rate 1/2 convolutional encoder and tabulate the corresponding encoded output pairs.
  - ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.
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**PROGRAM (MATLAB CODE)**

```
clc;

clear;

close all;

% Fixed input sequence
input_bits = [1 0 0 1 1];

% Convolutional encoder
trellis = poly2trellis(3,[7 5]);

encoded_bits = convenc(input_bits,trellis);

% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[]);

% Create output strings
for i = 1:length(input_bits)
    output_pair{i} = sprintf('%d%d', ...
        encoded_pairs(i,1),encoded_pairs(i,2));
end

% Create table
Bit_Position = (1:length(input_bits))';

Input_Bit = input_bits';
```

```

Output_Encoded = output_pair';
T = table(Bit_Position,Input_Bit,Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
% Plot
figure;
subplot(2,1,1)
stairs(input_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Input Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stairs(encoded_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Encoded Output Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on

```

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## PROCEDURE

### Objective (i): Convolutional Encoding

1. Open MATLAB and create a new script file.
2. Initialize the workspace using `clc`, `clear`, and `close all`.
3. Define the input bit sequence as: **[1 0 0 1 1]**.
4. Create a convolutional encoder trellis using constraint length **3** and generator polynomials
5. Encode the input sequence using the `convenc()` function.
6. Group the encoded bits into pairs corresponding to each input bit.
7. Create a table showing bit position, input bit, and encoded output pair.
8. Display and observe the encoded output pairs.

**Result:**

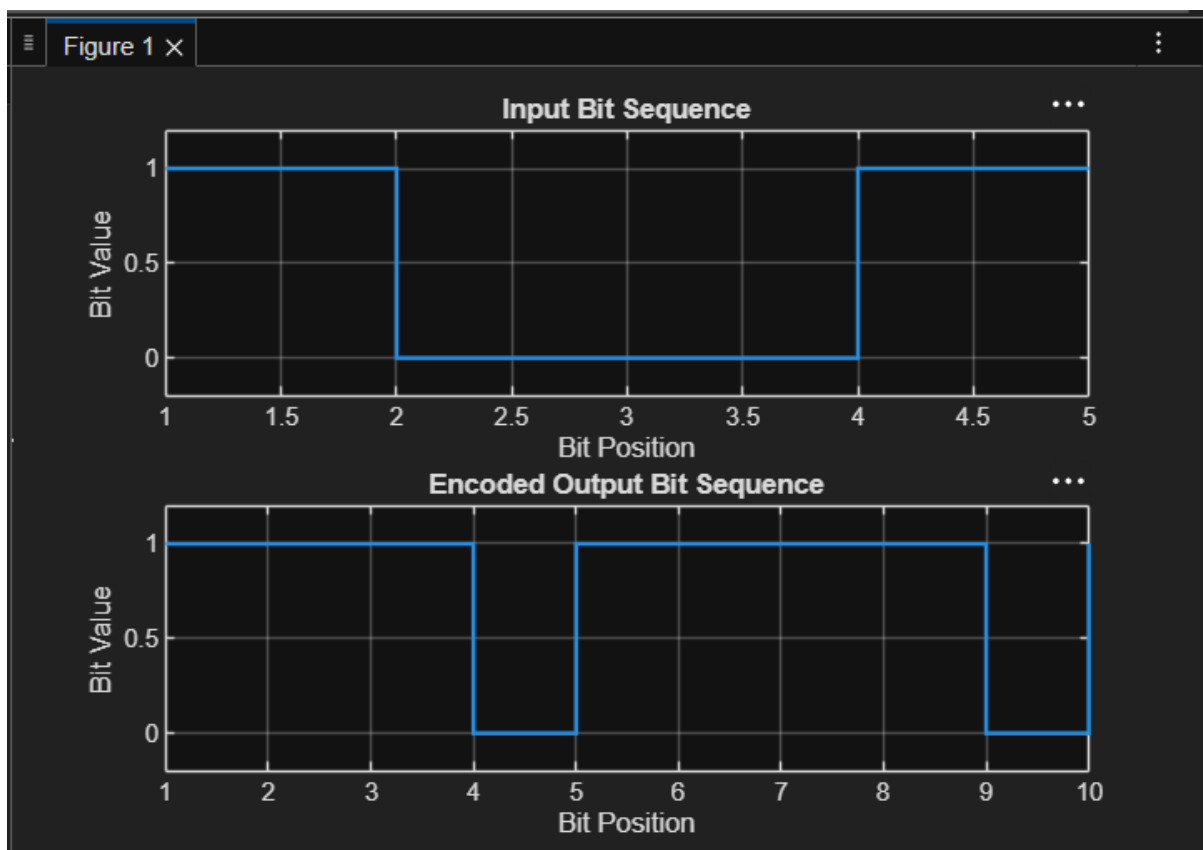
The convolutional encoder was successfully implemented using a constraint length of 3 and generator polynomials [7 5]. The input bit sequence [1 0 0 1 1] was encoded into the output sequence **1110111101**, and the corresponding encoded output pairs were tabulated successfully.

**Objective (ii): Plot and Analyze Sequences**

1. Create a figure window in MATLAB.
2. Plot the input bit sequence using the stairs() function in the first subplot.
3. Label axes and provide a suitable title.
4. Plot the encoded output bit sequence using the stairs() function in the second subplot.
5. Label axes and provide a suitable title.
6. Enable grid for clear visualization.
7. Compare the input and encoded sequences.
8. Observe redundancy introduced by encoding.

**Result:**

The input and encoded bit sequences were plotted using MATLAB. The encoded sequence contained **10 bits for 5 input bits**, confirming a **rate 1/2 convolutional code**. The plots clearly demonstrated the redundancy introduced by convolutional encoding, which enhances the error-correction capability of digital communication systems.

**OUTPUT (GRAPH);**

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## **FINAL RESULT**

The convolutional coding process was successfully simulated using MATLAB. The encoded output sequence and corresponding bit pairs were obtained and tabulated. The graphical comparison showed that the encoded sequence has increased length, confirming redundancy addition, which improves error detection and correction performance in communication systems.

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